Lectures 7–8: SCR and SCR Commutation

EEE 339 Power Electronics · Previous-year question solutions · Source: *class slides/Lecture 07 SCR.pdf* and *Lecture 08 SCR Commutation.pdf* (Rashid ch. 9)

Questions from these lectures (finals 2021 to 2024, term tests 2019 to 2020):

Asked in	Question	Marks	Solved in
2021 Q2(b), 2022 Q4(c), 2023 Q2(a), 2024 Q1(a)	Two-transistor model: turn-on condition / anode current / characteristics	6 to 8	Q1
2020 TT-1 Q2	I-V curve; holding and latching current	5	Q2
2021 TT-1 Q1	Five turn-on processes; effect of dv/dt with diagram	5	Q3
2022 Q4(d)	Methods of SCR turn-on and turn-off	8	Q4
2024 Q1(b)	Natural vs forced commutation, one device example each	6	Q5
2021 Q3(c)	Any one forced commutation circuit, with diagram and waveforms	6	Q6
2024 Q1(c)	Minimum gate pulse width (slide Example 14-1, word for word)	7	Q7
2024 Q1(d), 2024 Q3(b)	Why SCRs cannot be paralleled; second breakdown in thyristors	5, 7	Lecture 5 notes

The two-transistor model was asked in **all four** finals. Learn Q1 first.

Q1 · Two-transistor model of a thyristor

2021 Q2(b) [8] · 2023 Q2(a) [7]: Draw the two-transistor model of a thyristor and explain the turn-on condition with equations from this model.
2022 Q4(c) [6]: Using the two-transistor model explain the anode current in a thyristor. · 2024 Q1(a) [7]: With a two-transistor model explain the characteristics of an SCR.

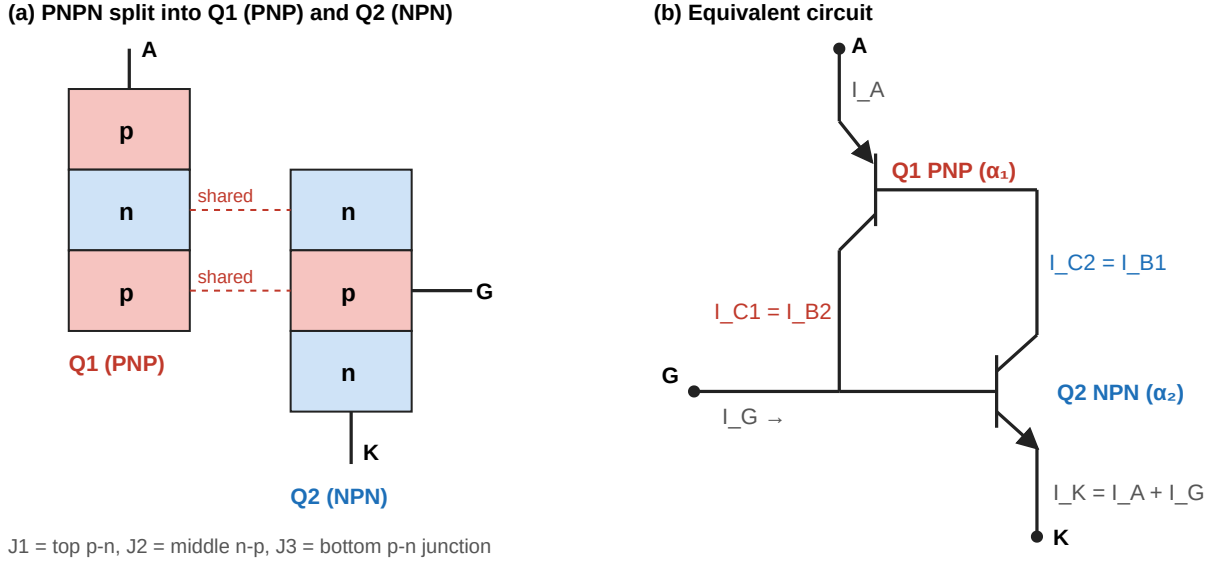


Fig. 1: (a) The PNPN structure viewed as two interconnected transistors. (b) Equivalent circuit.

Model. The regenerative (latching) action of a thyristor can be shown by splitting its PNPN structure into two complementary transistors: a PNP transistor **Q1** (p-n-p from the anode side) and an NPN transistor **Q2** (n-p-n from the cathode side). The middle n and p layers are shared: the **collector of each transistor drives the base of the other**, which is positive feedback.

Derivation. For any transistor, $I_C = \alpha I_E + I_{CBO}$, where α is the common-base current gain and I_{CBO} the collector-base leakage current.

For Q1 (emitter current = I_A): $I_{C1} = \alpha_1 I_A + I_{CBO1}$ ……(1)

For Q2 (emitter current = I_K): $I_{C2} = \alpha_2 I_K + I_{CBO2}$ ……(2)

Anode current (KCL at Q1): $I_A = I_{C1} + I_{C2}$ ……(3)

Cathode current: $I_K = I_A + I_G$ ……(4)

Put (1), (2), (4) into (3): $I_A = \alpha_1 I_A + I_{CBO1} + \alpha_2 (I_A + I_G) + I_{CBO2}$

$I_A(1 - \alpha_1 - \alpha_2) = \alpha_2 I_G + I_{CBO1} + I_{CBO2}$

$$\mathbf{I_A = (\alpha_2 I_G + I_{CBO1} + I_{CBO2}) / [1 - (\alpha_1 + \alpha_2)]} \quad \textbf{……(5)}$$

Turn-on condition (explaining equation 5).

- α_1 and α_2 are not constant: they are small at low emitter current and **increase as the current increases**.
- **Off state (forward blocking):** with $I_G = 0$ the currents are only leakage, so $\alpha_1 + \alpha_2$ is much less than 1 and $I_A \approx (I_{CBO1} + I_{CBO2})$, a tiny leakage current.
- **Gate current applied:** I_G increases I_K, so α_2 rises; I_{C2} rises, and since I_{C2} is the base current of Q1, I_{C1} rises; I_{C1} is the base current of Q2, so I_{C2} rises further. This is **regenerative** (positive feedback).
- As the currents grow, $\boldsymbol{\alpha_1 + \alpha_2 \to 1}$, the denominator of (5) $\to 0$, so $I_A \to \infty$ in theory. In practice I_A is limited only by the external circuit: **the thyristor has turned on (latched)**.
- Once I_A is above the latching current I_L, the loop sustains itself and the **gate signal can be removed**. The SCR stays on until I_A falls below the holding current I_H.
- Anything that raises α or the leakage currents (high temperature, high V_{AK} near V_{BO}, a large dv/dt, light) can also make $\alpha_1 + \alpha_2$ reach 1, which is why these are the other turn-on methods.

For 2024 Q1(a) ("explain the characteristics"): add Fig. 2 and link each region to the model: forward blocking ($\alpha_1 + \alpha_2 < 1$, only leakage); breakover at V_{BO} (avalanche leakage raises $\alpha_1 + \alpha_2$ to 1); a larger I_G reaches $\alpha_1 + \alpha_2 = 1$ at a lower voltage, so V_{BO} falls; on-state (latched, about 1 V drop); I_L and I_H; reverse blocking (J1 and J3 reverse-biased).

Q2 · 2020 TT-1: I-V curve; holding and latching current [5]

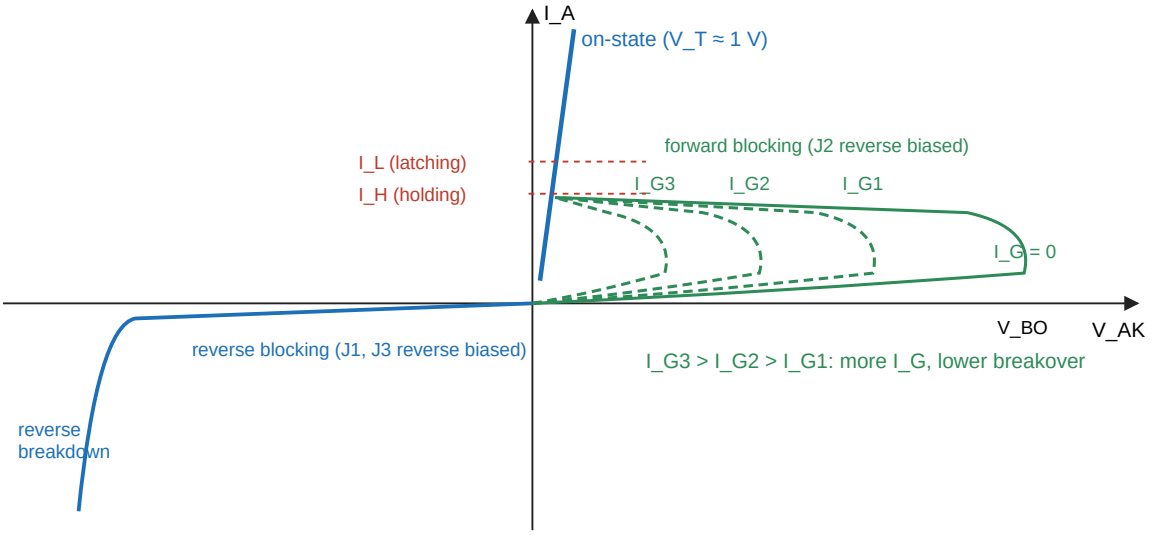


Fig. 2: SCR I-V characteristics with the effect of gate current.

- **Forward blocking:** anode positive; J1 and J3 forward-biased, J2 reverse-biased; only a small leakage (off-state) current flows.
- **Forward breakover V_BO:** if V_AK is raised far enough, J2 breaks down by avalanche; all three junctions conduct and the SCR switches to the **on-state** with only about 1 V drop. In practice V_AK is kept below V_BO and the SCR is turned on by gate current; more gate current means a lower breakover voltage.
- **Reverse blocking:** cathode positive; J1 and J3 reverse-biased (like two series diodes); only reverse leakage I_R flows, until reverse breakdown.

Latching current I_L: the minimum anode current required to keep the thyristor in the on-state **immediately after it has been turned on and the gate signal has been removed**.
Holding current I_H: the minimum anode current required to **maintain** the thyristor in the on-state. If I_A falls below I_H, a depletion region forms around J2 and the SCR returns to the blocking state. It is of the order of milliamperes.
I_L > I_H.

Q3 · 2021 TT-1: Five turn-on processes; effect of dv/dt [5]

Five ways to turn on a thyristor (slide):

1. **Thermal:** high temperature raises electron-hole pairs and leakage, so $\alpha_1 + \alpha_2$ rises toward 1. May cause thermal runaway; normally avoided.
2. **Light:** light striking the junctions creates electron-hole pairs (light-activated SCRs, LASCR).
3. **High voltage:** V_AK > V_BO; leakage starts regenerative turn-on. Can be destructive; avoided.
4. **dv/dt:** a fast-rising V_AK charges the junction capacitance (below).
5. **Gate current:** a positive gate-cathode voltage injects gate current; the **normal, practical method**. More gate current means a lower forward blocking voltage.

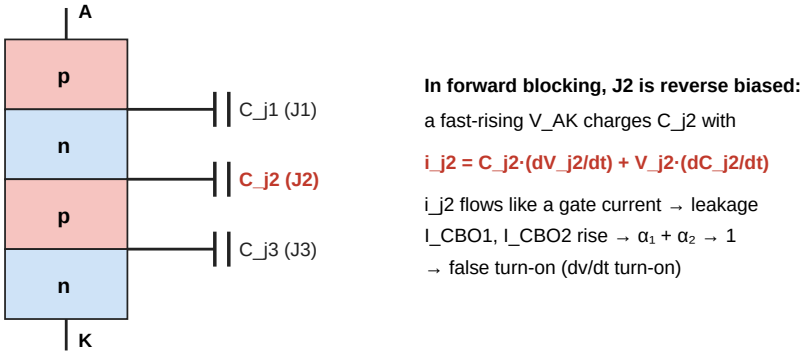


Fig. 3: Thyristor transient model: junction capacitances C_j1, C_j2, C_j3.

Effect of dv/dt: in forward blocking, J2 is reverse-biased and behaves like a capacitor C_j2. A rapidly rising anode voltage drives a charging current **i_j2 = C_j2·dV_j2/dt (+ V_j2·dC_j2/dt)** through it. This current acts like a gate current: it increases the leakage currents I_CBO1 and I_CBO2, so $\alpha_1 + \alpha_2$ can reach 1 and the SCR turns on **without a gate signal** (false triggering). A large charging current can also damage the device. Protection: an **RC snubber** across the SCR limits dv/dt; manufacturers specify the maximum allowable dv/dt.

Q4 · 2022 Q4(d): Methods of SCR turn-on and turn-off [8]

Turn-on methods

List and explain the five methods from Q3 (thermal, light, high voltage, dv/dt, gate current), stating that **gate triggering** is the practical method and the others are mostly unwanted. Add the dynamic turn-on picture:

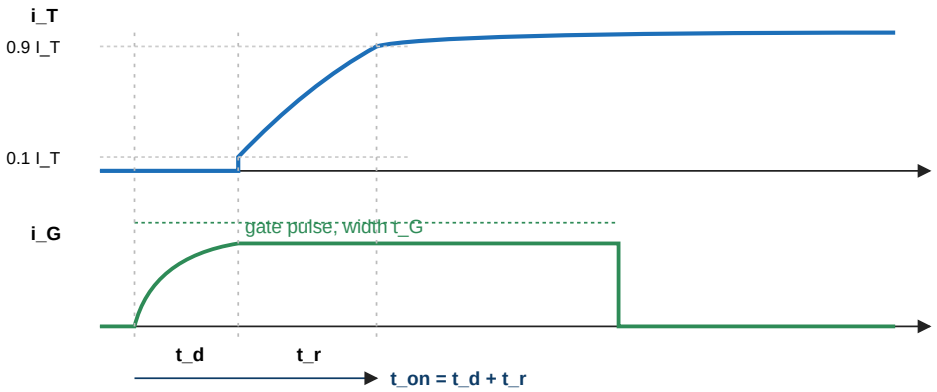


Fig. 4: Dynamic turn-on: after the gate pulse, the anode current needs the delay time t_d and rise time t_r.

Gate-pulse rules (slide): remove the gate signal once the SCR is on (to save gate loss); do not apply a gate signal while the SCR is reverse-biased; the gate pulse must last longer than the time the anode current needs to reach the holding (latching) value, and in practice longer than t_on.

Turn-off methods (commutation)

The gate cannot turn an SCR off. Turn-off (**commutation**) needs two conditions: (1) the anode current must be reduced below the holding current I_H, and (2) a reverse voltage must be applied for a sufficient time so that the excess carriers are removed and the SCR regains forward blocking.

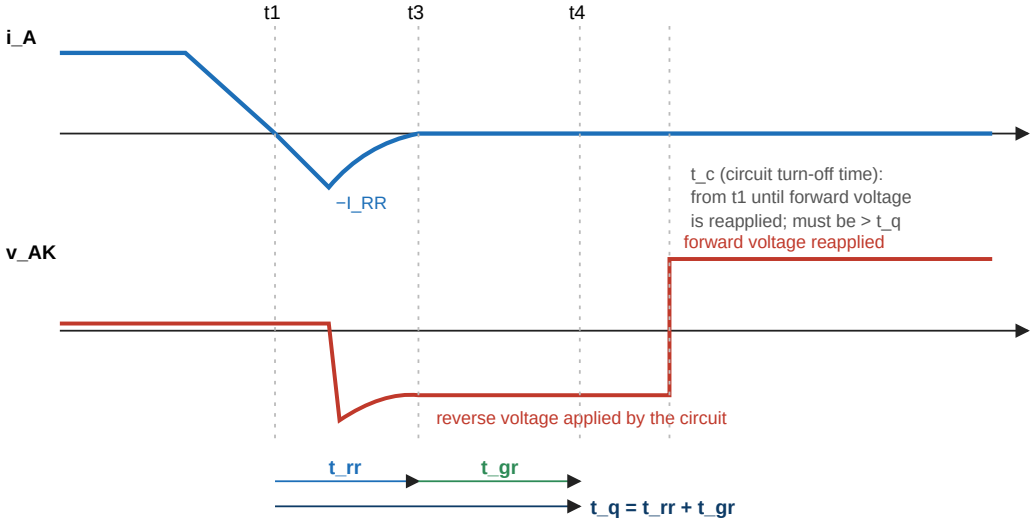


Fig. 5: Turn-off characteristics: t_q = t_rr + t_gr; the circuit must give t_c > t_q.

- **t_rr** (t1 to t3): reverse recovery current removes carriers from the outer junctions J1, J3.
- **t_gr** (t3 to t4): carriers at J2 disappear by recombination while reverse voltage is held. **t_q = t_rr + t_gr** (10 to 100 µs).
- The commutation circuit must give a **circuit turn-off time t_c > t_q**, or commutation fails.

Method	How the current is forced to zero
Natural (line, class F)	The AC supply itself: current passes through a natural zero at the end of the positive half cycle and the supply then reverse-biases the SCR.
Class A (self / resonant / load)	Load is an underdamped RLC circuit; the current oscillates to zero and the charged capacitor reverse-biases the SCR. Good above 1000 Hz.
Class B (self, resonant LC across SCR)	An LC circuit across the SCR (not in series with load) rings and turns the SCR off periodically; used in choppers.
Class C (complementary)	A second SCR in parallel; firing it puts a charged capacitor's reverse voltage across the main SCR. See Q6.
Class D (auxiliary / impulse)	An auxiliary SCR switches a pre-charged capacitor across the main SCR; diode D and L reverse the capacitor polarity.
Class E (external pulse)	A pulse transformer injects a reverse voltage pulse (lasting at least t_q) across the SCR.

Device grades (slide): converter-grade SCRs have t_q ≈ 50 to 100 µs (rectifiers, cycloconverters, AC regulators); inverter-grade SCRs have t_q ≈ 3 to 50 µs (choppers, inverters) and cost more.

Q5 · 2024 Q1(b): Natural vs forced commutation [6]

Q. Explain how natural commutation and forced commutation occur, giving one static power semiconductor device example for each.

Natural commutation (line / source / class F): the commutation voltage comes from the **AC supply itself**. At the end of every positive half cycle the anode current passes through a **natural current zero**, and immediately afterwards the supply applies a **reverse voltage** across the SCR. Both turn-off conditions are met, so the SCR turns off by itself.

Example: an **SCR (thyristor)** in a single-phase phase-controlled rectifier. Natural commutation is also used in AC voltage controllers (SCR or TRIAC), cycloconverters and line-commutated inverters.

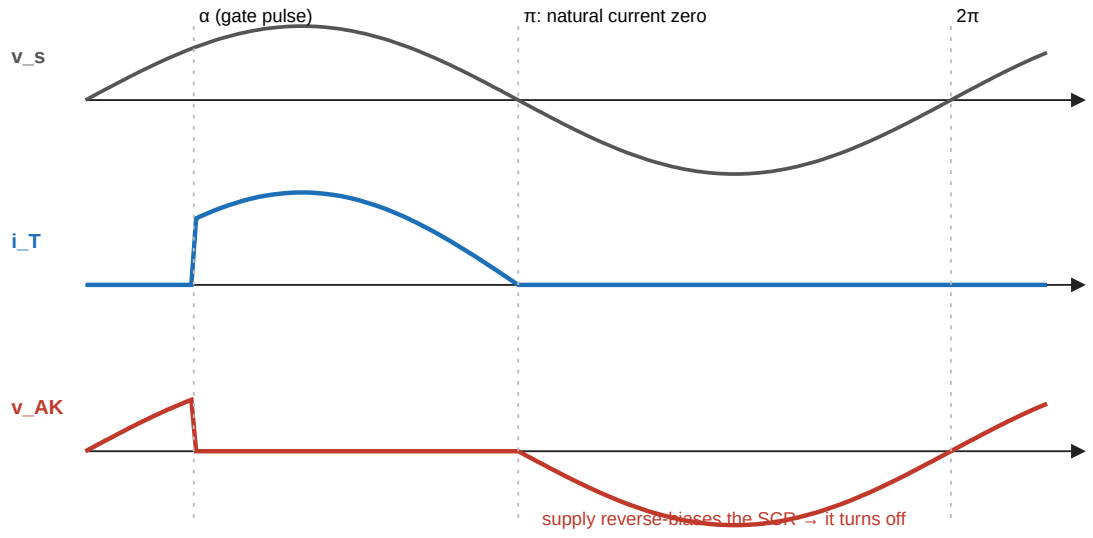


Fig. 6: Natural commutation of an SCR (resistive load): the current ends at π and the supply then reverse-biases the SCR.

Forced commutation: in **DC circuits** there is no natural current zero. An external **commutation circuit of L and C** forces the anode current to zero and applies a reverse voltage across the SCR (Classes A to E).

Example: an **SCR in a DC chopper** (DC-DC converter) or in a forced-commutated inverter.

	Natural commutation	**Forced commutation**
Supply	AC	DC
Source of commutation voltage	The supply itself	External L-C commutation circuit
Extra components	None	Inductors, capacitors, auxiliary SCRs
Cost / losses	Low	Higher
Also called	Line, source, class F	Classes A, B, C, D, E
Device example	SCR in a controlled rectifier	SCR in a DC chopper

Q6 · 2021 Q3(c): One forced commutation circuit [6]

Q. Briefly describe any one kind of thyristor forced commutation circuit with necessary diagram and waveforms.

Chosen: Class C (complementary) commutation. The main SCR T_1 is in series with the load R_1; an auxiliary (complementary) SCR T_2 with R_2 is connected in parallel with it, and a capacitor C joins the two anodes.

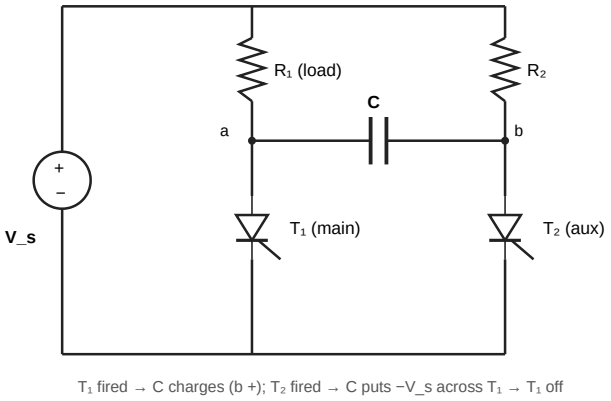


Fig. 7a: Class C commutation circuit.

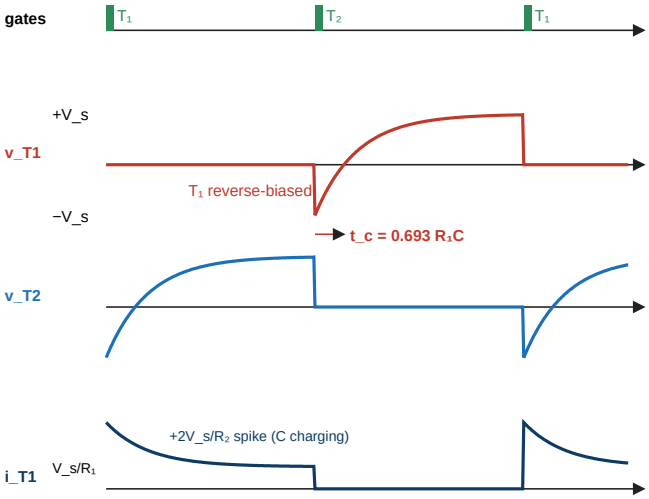


Fig. 7b: Waveforms ($R_1 = R_2$, steady state).

1. **T_1 fired:** load current V_s/R_1 flows through R_1 and T_1. At the same time C charges through R_2 and T_1 until plate b is at +V_s (a at 0). i_T1 has an extra decaying capacitor-charging spike.
2. **T_2 fired (to turn T_1 off):** point b drops to 0. Because a capacitor's voltage cannot change instantly, point a jumps to **−V_s**: the charged capacitor puts a **reverse voltage across T_1**, its current falls to zero and it turns off.
3. C now charges through R_1 (and T_2) in the opposite direction: v_T1 = V_s(1 − 2e^(−t/R_1C)). T_1 stays reverse-biased until v_T1 crosses zero, which gives the **circuit turn-off time**
   `t_c = R₁C·ln 2 = 0.693 R₁C` `(must be > t_q of T₁)`
4. **T_1 fired again:** a drops to 0, b jumps to −V_s, so T_2 is reverse-biased and turns off. The cycle repeats; each SCR turns the other off.

Features (slide): also called complementary commutation; the SCR turns off by the reverse voltage of a charged capacitor; very reliable; useful even below 1000 Hz; used in single-phase inverters with centre-tapped transformers.

Q7 · 2024 Q1(c): Minimum gate pulse width [7]

Q. The holding current of the thyristors in a single-phase full converter is I_H = 500 mA and the delay time is t_d = 1.5 µs. The converter is supplied from a 120 V, 60 Hz supply and has a load of L = 10 mH and R = 10 Ω. The converter is operated with a delay angle α = 30°. Determine the minimum value of gate pulse width t_G.

This is **Example 14-1 on your Lecture 8 slides**, word for word.

Step 1: supply voltage at the firing instant. $V_m = \sqrt{2} \times 120 = 169.7$ V; at $\omega t = \alpha = 30°$:
$V_1 = V_m \sin \alpha = 169.7 \times \sin 30° =$ **84.85 V**

Step 2: initial rate of rise of anode current. Just after triggering the inductor limits the current ($R \cdot i$ is still small):
$di/dt = V_1/L = 84.85 / (10 \times 10^{-3}) =$ **8485 A/s**

Step 3: time for the current to reach the holding value. Taking di/dt as constant for this short time:
$t_1 = I_H / (di/dt) = 0.5 / 8485 =$ **58.93 µs**

Step 4: minimum gate pulse width. The gate must stay on during the delay time plus this rise time:
$t_G = t_1 + t_d = 58.93 + 1.5 =$ **60.43 µs**

Answer: $t_{G(min)} \approx$ 60.43 µs (this is the slide's answer)

Check with the exact RL current $i(t) = (V_1/R)(1 - e^{(-tR/L)})$: $t_1 = -(L/R)\ln(1 - I_H R/V_1) = 60.73$ µs, so $t_G = 62.23$ µs. Both are accepted; write the slide method and mention the assumption "di/dt constant". In practice t_G is made longer than t_{on}.

Q8 · Related questions answered in your Lecture 5 notes

- **2024 Q1(d):** why SCRs (and BJTs) cannot be paralleled easily: negative temperature coefficient, unequal on-state drops and turn-on times, so one device hogs the current. Needs series R/L and simultaneous strong gate pulses.
- **2024 Q3(b):** secondary breakdown occurs in thyristors because they are bipolar devices with negative temperature coefficient and regenerative action: current filaments form hot spots.

Likely new questions (short answers)

Predicted **Q. Explain the dynamic turn-on and turn-off characteristics of an SCR.**

Turn-on (Fig. 4): $t_{on} = t_d + t_r$ (t_d: anode current rises to 0.1 I_T; t_r: 0.1 to 0.9 I_T). Turn-off (Fig. 5): $t_q = t_{rr} + t_{gr}$; the circuit turn-off time t_c must exceed t_q.

Predicted **Q. Distinguish converter-grade and inverter-grade SCRs. / Compare thyristors and transistors.**

Converter grade: slow, $t_q \approx 50$ to 100 µs, cheaper; rectifiers, cycloconverters, AC regulators. Inverter grade: fast, $t_q \approx 3$ to 50 µs, costlier; choppers, inverters. Thyristors vs transistors (slide): thyristors have lower conduction loss and higher power capability; transistors switch faster with lower switching loss.

Predicted **Q. Describe Class A / Class B / Class D / Class E commutation.**

Use the one-line descriptions in the Q4 table. Class A: underdamped RLC load gives a natural current zero (high frequency). Class B: LC across the SCR, periodic on/off (choppers). Class D: auxiliary SCR switches a charged capacitor; the diode prevents reverse discharge. Class E: pulse transformer applies an external reverse pulse at least t_q long.